

Web Appendix:

Consumers Undervalue Multi-Option Alternatives in Two-Stage Choice

Web Appendix A: Main Effect, Conditioned on Relative Ratings of S, M_H , and M_L

Web Appendix B: Study 1b

Web Appendix C: Study 1c

Web Appendix D: Study 1d

Web Appendix E: Study 2b

Web Appendix F: Additional detail on Study 3

Web Appendix G: Study 3b

Web Appendix H: Study 5 Trial Creation

Web Appendix I: Additional detail on Study 6

Web Appendix J: Study 6b

Web Appendix K: Study 6c

Web Appendix L: Study 6d

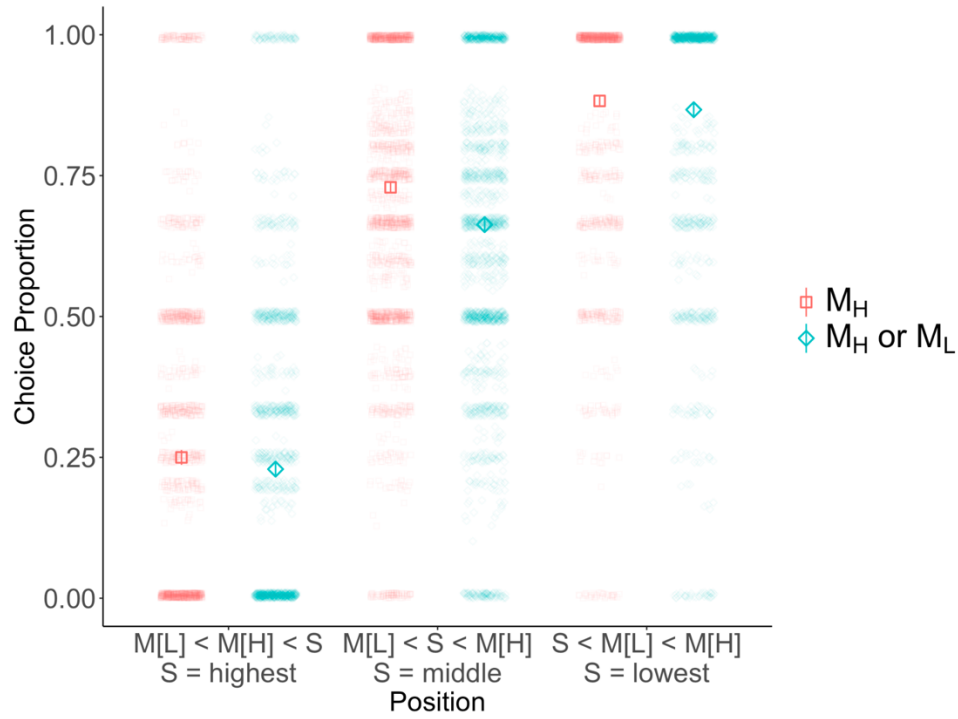
Web Appendix M: Study 6e

Web Appendix N: Extra detail on Heterogeneity Simulations

Web Appendix O: Supplementary Tables

Appendix A: Main Effect, Conditioned on Relative Ratings of S, M_H , and M_L

Figure S1: Undervaluation by Relative Ratings



Notes: Main effect of undervaluation, separated by the relative rating of S (compared to M_H and M_L). In all situations, participants are less likely to choose M_{HL} than they are to choose M_H when compared to the same single-option alternative S (S = highest: $t(2401) = 3.66, p < .001$; S = middle: $t(2841) = 12.62, p < .001$; S = lowest: $t(2113) = 3.19, p = .001$). Data is collapsed across the movie studies (Studies 1, 1b, 1c, 2, 2b, 3, and 3b), which have individual ratings per option. Data from Studies 3 and 3b rely on 1v2 comparisons. Bars indicate s.e.m. across participants.

Appendix B: Study 1b

Method

Participants. For this preregistered experiment, we collected responses from 604 Amazon Mechanical Turk workers; each earned \$1.25 for participating.

Materials and Procedure. The materials and procedure were identical to Study 1.

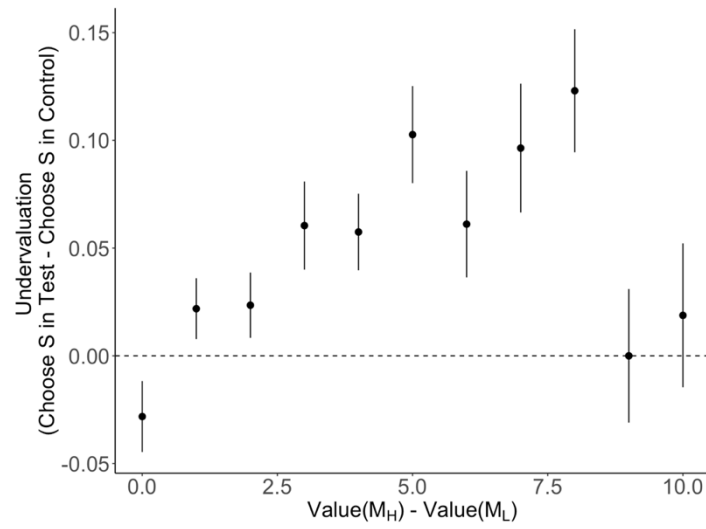
Exclusions and Data Preprocessing. We followed the same exclusion rule used in Study 1. These criteria resulted in the exclusion of 80 participants, leaving us with a sample size of 524.

Results

Using the same approach as in Study 1, we found evidence for undervaluation, $M = 0.04$, 95% CI = [0.03, 0.05], $t(523) = 6.75$, $p < .001$. Using that approach, we find that 41% of participants exhibited undervaluation, with 37% exhibiting no difference and 22% exhibiting an effect in the opposite direction. As in Study 1, we find significant evidence for an effect of M_H - M_L preference strength on undervaluation, $b = 0.011$, $SE = 0.002$, $p < .001$ (Fig. S2).¹

¹ Due to a since-addressed coding error in an early analysis of Study 1, we collected this dataset with additional preregistered exclusions. We specified that we would remove all choices where the M_H - M_L difference was less than 5. With that exclusion, we find a larger effect, $M = 0.09$, 95% CI = [0.06, 0.12], $t(335) = 6.64$, $p < .001$.

Figure S2: Preference Strength and Undervaluation



Notes: Relationship between ($M_H - M_L$) preference strength and degree of undervaluation. Undervaluation increases as M_H gets progressively better than M_L . Analysis reported in text controls for rating of S and sum of ratings of M_H and M_L . Bars represent s.e.m. across participants.

Appendix C: Study 1c

Method

Participants. For this preregistered experiment, we collected responses from 499 Amazon Mechanical Turk workers; each earned \$1.25 for participating.

Materials and Procedure. The materials and procedure were identical to Study 1, with the following change: instead of a multiple-choice comprehension question about multi-option alternatives, participants were asked to consider a choice between two theaters (one showing a single movie, *Beauty and the Beast*, and one showing two movies, *Fantasia* and *Jurassic World*). Participants then encountered the following prompt: “In the example decision above, suppose you picked theater 2 (*Fantasia*; *Jurassic World*). Please briefly describe how which movie you watch would be determined.”

As in Study 1, participants also responded to comprehension questions to ensure that they understood (1) that the choices were hypothetical and (2) that their choices would not influence the number of choices that they would have to make.

Exclusions and Data Preprocessing. We used the same exclusion criteria as in Study 1: participants had to rate at least 20 films and their filler choices had to be directionally predicted by their ratings. Furthermore, as preregistered, we only included participants in analysis if two coders (blind to hypothesis and choice data) agreed that the participants’ answer to the open-ended comprehension question unambiguously indicated that the participant would watch their

choice of movie from the pair. In total, we excluded 187 participants from the original sample, resulting in a final sample size of $N = 312$.

Results

Using the same approach as in Study 1, we found evidence for undervaluation, $M = 0.02$, 95% CI = [0.01, 0.04], $t(311) = 3.18$, $p = .002$. Using that approach, we find that 35% of participants exhibited undervaluation, with 43% exhibiting no difference and 22% exhibiting an effect in the opposite direction. As in Study 1, we find significant evidence for an effect of M_H - M_L preference strength on undervaluation, $b = 0.017$, $SE = 0.002$, $p < .001$.

Appendix D: Study 1d

Method

Participants. For this preregistered experiment, we collected responses from 388 Amazon Mechanical Turk workers; each earned \$1.25 for participating.

Materials and Procedure. The materials and procedure were identical to Study 1, with the following change: instead of choosing between two theaters, participants chose between two trains, each of which went to a different part of town. Each part of town had either one or two theaters, each of which was showing one film.

As in Study 1, participants also responded to comprehension questions to ensure that they understood (1) that the choices were hypothetical and (2) that their choices would not influence the number of choices that they would have to make.

Exclusions and Data Preprocessing. We used the same exclusion criteria as in Study 1: participants had to rate at least 20 films and their filler choices had to be directionally predicted by their ratings. In total, we excluded 38 participants from the original sample, resulting in a final sample size of $N = 350$.

Results

Using the same approach as in Study 1, we found evidence for undervaluation, $M = 0.02$, 95% CI = [0.01, 0.04], $t(349) = 2.84$, $p = .004$. Using that approach, we find that 39% of participants exhibited undervaluation, with 34% exhibiting no difference and 27% exhibiting an effect in the opposite direction. As in Study 1, we find significant evidence for an effect of M_H - M_L preference strength on undervaluation, $b = 0.012$, $SE = 0.003$, $p < .001$.

Appendix E: Study 2b

Method

Participants. For this preregistered experiment, we collected responses from 201 Amazon Mechanical Turk (AMT) workers. They earned \$1.50 for their participation.

Materials and Procedure. The procedure for this experiment was identical to Study 2.

Exclusions and Data Preprocessing. As specified in our preregistration, we excluded anyone who failed to rate at least 20 films. We also excluded anyone whose filler choices were not directionally predicted by their ratings in a logistic regression of *ChooseLeft* on (*RatingLeft* - *RatingRight*). These criteria resulted in the exclusion of 24 participants, leaving us with a final sample size of 177. As specified in our preregistration, we excluded any test-control choice pairs that were generated from unrated films.

Results

As in previous experiments, we find evidence of undervaluation, $M = 0.03$, 95% CI = [0.01, 0.04], $t(176) = 2.94$, $p = .003$. Next, we make several comparisons between ratings. First, we look at whether the rating of S changes, depending on whether it is part of a test choice or a control choice, and find that it was rated marginally significantly lower following a control choice than a test choice ($M = -0.45$, 95% CI = [-0.93, 0.04], $t(175) = 0.069$). Next, we make

within-subject comparisons among three values: (i) the rating of M_H following control choices, (ii) the rating of M_H following test choices and (iii) the rating of M_{HL} following test choices, using one-sample t-tests.

To reduce sampling variability in the selected movies we subtracted out the initial ratings provided by participants, as in Study 2. We find that M_{HL} is rated lower following test than is M_H following control ($M = -0.67$, 95% CI = $[-1.00, -0.34]$, $t(174) = 4.01$, $p < .001$). M_H is also rated marginally significantly lower following test than following control ($M = -0.28$, 95% CI = $[-0.57, .005]$, $t(174) = 1.94$, $p = .054$). M_{HL} is rated significantly lower following test than M_H is following test ($M = -0.37$, 95% CI = $[-0.73, -0.01]$, $t(174) = 2.02$, $p = .044$).

Table S1

Mean Ratings of M_H and M_{HL}

Comparison	M_H (control)	M_H (test)	M_{HL} (test)
Mean Rating (SE)	6.54 (0.20)	6.25 (0.21)	5.94 (0.21)
Mean (Rating – Initial Rating) (SE)	0.21 (0.13)	-0.06 (0.13)	-0.48 (0.17)

Notes: For the bottom row, we first subtract (from each post-choice rating) the rating of movie M_H provided by the participant during the first task in the study.

In our preregistration for this study (which was collected before Study 2), we did not specify whether we would control for the initial ratings of the films. When we do not control for them, we find that the rating of M_{HL} following test choices is significantly lower than the rating of M_H following control choices ($M = -0.60$, 95% CI = $[-1.06, -0.14]$, $t(175) = 2.59$, $p = .010$) and that the rating of M_H following test choices is not significantly lower than the rating of M_H following control choices ($M = -0.29$, 95% CI = $[-0.73, 0.14]$, $t(175) = -1.33$, $p = .186$). The rating of M_{HL} following test choices is not significantly lower than the rating of M_H following

test choices ($M = -0.31$, 95% CI = $[-0.74, 0.13]$, $t(175) = -1.39$, $p = 0.166$).

In sum, it is clear that M_{HL} is rated lower following test than M_H is following control choices. Given the most powerful test (accounting for sampling variability of differences in movies), M_{HL} is rated lower following test than M_H is following test, suggesting that part of the undervaluation effect is indeed due to undervaluation of the holistic option, not just the more-preferred option in the presence of the other option.

Appendix F: Additional Detail on Study 3

Filler trials in Study 3

Among the 17 filler trials, 9 were simple binary trials, each of which comprised two single-movie theaters. The simple binary trial films were randomly selected from the rated films (independently from the selection process for the test trials, above). If a participant rated fewer than 18 films, then we sampled from the unrated films when necessary. These trials were used for our preregistered exclusion criteria and to add some variability to the choices being made by participants. The remaining 8 filler trials were designed to look like the trials from the other trial categories specified above. The films in each trial were randomly selected from the rated films (separately from the selection process mentioned above).

Trials were presented in a randomized-block order. Participants completed 3 or 4 filler questions before each block of 5 choices. Each block contained one question of each type (1v1, 1v2, 1v3, 2v2, 2v3), each from a different matched set. The display order of the sub-options (e.g., M_A -or- M_B vs. M_B -or- M_A) was randomly selected at the participant-by-trial level.

Appendix G: Study 3b

Method

Participants. For this preregistered experiment, we collected responses from 517 AMT workers. They earned \$1.50 for their participation.

Materials and Procedure. The procedure was identical to Study 3.

Exclusions and Data Preprocessing. As specified in our preregistration, we excluded anyone who failed to rate at least 18 films, as they would have fewer usable primary choices and thus noisier estimates. We also excluded anyone whose simple binary choices were not directionally predicted by their ratings in a logistic regression of *ChooseLeft* on (*RatingLeft-RatingRight*) as that indicates lack of minimal attention during ratings, choice, or both. These criteria resulted in the exclusion of 158 participants, leaving us with a final sample size of 359.

Also in line with our preregistration, we excluded any choice sets that were generated from unrated films.

Results

In this study, we can test for undervaluation of several theaters. We first present analyses using the Study 3 criterion of only using sets in which the added movie was rated at least 2 points lower than the existing sub-options. First, analogous to analyses in Studies 1, 1b, 2, and

2b, we can compare choices in 1v1 and 1v2 trials to test for undervaluation of the 2-movie theater in 1v2 choices. To test for undervaluation in this case, for every participant, we calculated $S_A > M_{AB}$ as (Chose S in 1v2 choices / number of valid 1v2 choices). We also calculated $S_A > M_A$ as (Chose S in 1v1 choices / number of valid 1v1 choices). We tested $(S_A > M_{AB}) - (S_A > M_A)$ using a one-sample t-test and found evidence for undervaluation, $M = 0.05$, 95% CI = [0.03, 0.07], $t(655) = 4.95$, $p < .001$. In other words, choice of M_A was 5 percentage points higher than choice of M_{AB} .

We used the analogous approach to compare 1v1 against 1v3, 1v2 against 2v2, and 2v2 against 2v3. In each case, we find undervaluation (1v1 vs. 1v2 trials: $M = 0.08$, 95% CI = [0.04, 0.12], $t(283) = 3.83$, $p < .001$; 1v1 vs. 1v3 trials: $M = 0.08$, 95% CI = [0.03, 0.12], $t(224) = 3.23$, $p = .001$; 1v2 vs. 2v2 trials: $M = 0.07$, 95% CI = [0.03, 0.12], $t(270) = 3.02$, $p = .003$; 2v2 vs. 2v3 trials: $M = 0.06$, 95% CI = [0.01, 0.12], $t(180) = 2.16$, $p = .03$).

For this experiment (which was collected prior to Study 3), we did not preregister the requirement that the added sub-option be rated at least two points lower than the existing sub-option(s), as we did in Study 3. Without that requirement, only the first two comparisons are significant: 1v1 vs. 1v2 trials: $M = 0.04$, 95% CI = [0.01, 0.06], $t(358) = 2.95$, $p = .003$; 1v1 vs. 1v3 trials: $M = 0.02$, 95% CI = [-0.002, 0.05], $t(358) = 1.80$, $p = .07$; 1v2 vs. 2v2 trials: $M = 0.004$, 95% CI = [-0.02, 0.03], $t(358) = 0.37$, $p = .72$; 2v2 vs. 2v3 trials: $M = -0.03$, 95% CI = [-0.06, 0.01], $t(345) = -1.62$, $p = .11$.

Appendix H: Study 5 Trial Creation

In Study 5, we designed the trials in a very similar way to Study 4. We first collected pilot data, where we asked a separate group of participants to rank-order the video categories (i.e., Animals, Travel, Beauty/Fashion, Gaming, Music Performance, and Cooking). Then, these pilot participants rated all 48 videos (represented by a screenshot and caption) on how much they would want to watch them.

Based on this pilot data, we imputed preference orderings for Study 5 participants for our test and control trials. Specifically, we identified 5 high-value, 5 medium-value, and 5 low-value videos for each Study 5 participant, based on their rank-ordering of the six video categories. We identified them using the following system. For the participant's highest-rated category, we looked at the average pilot ratings for each video in that category, among pilot participants who also ranked that category highest. We used the three highest-rated videos in that category. For the participant's 2nd-highest-rated category, we looked at the average pilot ratings for each video in that category, among pilot participants who ranked that category 2nd-highest. We used the two highest-rated videos in that category. We used the same process to select the 5 medium-value videos (the 5th- and 6th-highest rated videos in the 3rd-highest category, and the 2nd-, 3rd-, and 4th-highest rated videos in the 4th-highest category) and the 5 low-value videos (the 7th- and 8th-highest rated videos in the 5th-highest category, and the 6th-, 7th-, and 8th-highest rated videos in the lowest category). To create each of our five paired test and control trials, we randomly selected (without replacement) one video from each value level (high/medium/low). The high-value video was assigned to be M_H , the medium-value video was assigned to be S , and the low-value video was assigned to be M_L . For a full list of our stimuli (and the average rating order within categories), please see our ResearchBox.

The filler trials were created using randomly-selected videos from the set not listed above.

Appendix I: Additional Details on Study 6

Trial Generation Process

To generate the 34 trials, we started with the set of all possible events, separated by domain (i.e., cards, dice, and coins) and probability. In the card domain, we only used suits/colors (e.g., drawing a black card, drawing a spade, drawing a red card or a club). In the dice domain, we used every combination of numbers (e.g., rolling a 1, rolling a 1 or 2 or 4 or 6, rolling a 2 or 3 or 5). In the coin domain, we used every outcome of flipping one or two coins (e.g., flipping heads on one coin, flipping tails on two coins, flipping at least one head on two coins). We then randomly selected, for each participant and with replacement, three domains and three probabilities within those domains to fill the roles of S, M_H , and M_L . The only constraint we placed on the assignment of domains and probabilities to the three roles (S, M_H , and M_L) was to require that the probability of M_H be greater than the probability of M_L , and the probabilities of S and M_H to be different. If this was not possible (e.g., if the sampled probabilities were equal), then we resampled all three roles. Once the domains and probabilities were established, to populate control and test choices, we randomly sampled with replacement three events for S, three events for M_H , and two events for M_L . For instance, if the domain for S was dice and the probability for S was 0.5, then we sampled three events (with replacement) from the entire list of possible dice events with probability 0.5 (e.g., rolling a 1 or 2 or 3; rolling a 3 or 5 or 6; rolling a 1 or 3 or 4). To determine the monetary amounts for each option, we randomly selected two dollar amounts ranging from \$2 to \$10; the lower amount was assigned to whichever of {S, M_H } had a higher probability, and higher amount was assigned to whichever of {S, M_H } had a lower

probability; this approach ensured a tradeoff between S and M_H . In all of the gambles studies, the monetary amounts for M_H and M_L were equal.

At the end of the survey, participants also completed a corresponding M_H vs. M_L trial from one of the 10 trial sets that they completed. Three attention check trials were used as well. These attention check trials were all binary choices. There were three structures; we included on trial of each: (i) $\text{prob}(A) = \text{prob}(B)$ and $\text{dollar}(A) > \text{dollar}(B)$; (ii) $\text{prob}(A) > \text{prob}(B)$ and $\text{dollar}(A) = \text{dollar}(B)$; (iii) $\text{prob}(A) > \text{prob}(B)$ and $\text{dollar}(A) > \text{dollar}(B)$.

Attention Manipulation

Participants completed two within-subject conditions: pro-undervaluation and anti-undervaluation. The control trials ($n = 10$) had the same structure in both conditions. Specifically, both S and M_H were shown three times each in an alternating sequence. Each option was shown once for 1.75 seconds, once for 1.575 seconds, and once for 1.4 seconds. These durations appeared in a randomized order within each option.

Pro-undervaluation. In the pro-undervaluation test trials ($n = 5$), S was visible for the same amount of time as in the control trials: once for 1.75 seconds, once for 1.575 seconds, and once for 1.4 seconds. M_H was only shown twice: once for 1.75 seconds, and once for 1.4 seconds. M_L was shown three times: once for 2.275 seconds, once for 2.1 seconds, and once for 1.925 seconds. Moreover, we required M_L to be shown first and last. The order of all other information reveals was randomized, such that an option never appeared twice in a row.

Anti-undervaluation. In the pro-undervaluation test trials ($n = 5$), S was visible for the same amount of time as in the control trials: once for 1.75 seconds, once for 1.575 seconds, and once for 1.4 seconds. M_L was only shown twice: once for 1.75 seconds, and once for 1.4 seconds. M_H was shown three times: once for 2.275 seconds, once for 2.1 seconds, and once for 1.925 seconds. Moreover, we required M_H to be shown first and last. The order of all other information reveals was randomized, such that an option never appeared twice in a row.

Appendix J: Study 6b

In Study 6b, we replicate the correlational attention results of Study 5 in the domain of gambles. We measure the information acquisition process during all choices and connect it to our main effect. We also manipulated whether the M_H vs. M_L dominance relationship was high-transparency (easy to identify) or low-transparency (difficult to identify). The monetary outcomes and the probabilities of winning were held constant between high-transparency pairings and low-transparency pairings, but in the high-transparency pairings, M_H always contained the event M_L and in low-transparency pairings, M_H never contained the event M_L . An example of a high-transparency pairing is M_H = “\$2 if you roll a 1 or 2 or 3” and M_L = “\$2 if you roll a 1.” An example of a low-transparency pairing is M_H = “\$2 if you draw a black card” and M_L = “\$2 if you roll a 4.” We hypothesized that a more (vs. less) transparent dominance relationship would lead to less undervaluation as the implications for the second-stage choice of M_H vs. M_L are clearer for more-transparent pairings. If participants can more easily “edit out” or “ignore” the inferior sub-option, then we may see this reflected in their cursor movements. More specifically, we might expect that they spend less time on M_L in high-transparency (vs. low-transparency) trials.

Method

Participants. For this preregistered study, we collected responses from 302 AMT workers. They earned \$2.50 (the first 20 participants) or \$2.75 (the remaining 282 participants) for their participation. Five randomly-selected participants received the outcome of one of their decisions, as detailed below.

Materials and Procedure. This study was similar to Study 6, with the following changes. First, the trials were not randomly generated at the subject-level and instead came from a predetermined set, as specified in our ResearchBox. Second, the M_H vs. M_L dominance relationship was manipulated to be either high- or low-transparency. Third, the information about a gamble (i.e., the details of S , M_H , and/or M_L) was not visible unless the participant hovered their cursor over each option. Fourth, we tracked participants' mouse movements, rather than showing them the information in a prespecified order. Specifically, while participants made their choices, we recorded the order in which participants viewed each piece of information (e.g., S , M_H , and M_L) and the durations of these information acquisitions in a MouseLab-like paradigm (Johnson et al. 1989). We measured (separately) attention to S , M_H , and M_L . Participants made a total of 65 incentivized choices. These choices included: (1) high-transparency test choices: S vs. M_{HL} (HT), (2) low-transparency test choices: S vs. M_{HL} (LT), (3) high-transparency control choices: S vs. M_H (HT), (4) low-transparency control choices: S vs. M_H (LT), (5) trinary choices: S vs. M_H vs. M_L , and (6) binary choices between M_H and M_L . At the end of the survey, participants rank-ordered options within 10 $\{S, M_H, M_L\}$ triplets. See below for more specific trial information. In this study, all five randomly-selected participants won money (\$3, \$5, \$4, \$2 and \$4).

As in the previous studies, participants completed several comprehension check questions, including a question about the meaning of the test choices, i.e., that choosing M_{HL} implied that they would get to choose an alternative from the set M_{HL} . They were required to get all questions correct before moving on to the choices.

Exclusions and Data Preprocessing. As specified in our preregistration, we excluded anyone who picked the obviously dominated option in any of the three attention-check questions (described in Appendix H) and we excluded anyone who did not mouse-over the boxes in the instructions as instructed. This resulted in the exclusion of 93 participants, leaving us with a sample size of 209.

For the mouse-tracking data, we converted the hover-times (i.e., the times that participants spent hovering over the available information) into proportions from 0 to 1 at the trial level.

Trial Generation Process. In this experiment, participants made 65 incentivized choices. Some were control binary choices (e.g., S vs. M_H), some were trinary choices (e.g., S vs. M_H vs. M_L) and some were test choices (e.g., S vs. M_{HL}). For each of 10 trial sets t , a given participant completed the following trials: (1) low transparency: S^{t1} vs. M_{HL}^{t1} (L), where M_H^{t1} and M_L^{t1} use different mechanisms; (2) high transparency: S^{t2} vs. M_{HL}^{t2} (H), where M_H^{t2} and M_L^{t2} use the same mechanism; (3) trinary: S^{t3} vs. M_H^{t3} vs. M_L^{t3} ; (4) control 1: binary choice between S^{t3} and M_H^{t3} , where the mechanism (card draw, die roll, or coin flip) for M_H^{t3} is the same as the mechanism for M_H^{t1} . If the mechanisms for M_H^{t1} and M_H^{t2} were different, then the participant completed a fifth trial for that set: (5) control 2: S^{t3} vs. M_H^{t3} , where the mechanism for M_H^{t3} is the same as the mechanism for M_H^{t2} .

Participants completed all 10 trial sets, as specified in the previous paragraph. At the end of the survey, participants completed a rank-ordering choice (of S, M_H , and M_L) within each set with the same mechanisms as (1), above. They also completed the two corresponding M_H vs. M_L trials (i.e., M_H^{t1} vs. M_L^{t1} and M_H^{t2} vs. M_L^{t2}) from one of the trial sets that they completed. As in

Study 6, at the end of data collection, we randomly selected five participants and then randomly selected one of their choices to play out. In this study, all five participants won money (\$3, \$5, \$4, \$2 and \$4).

Results

Choice Results. We tested for undervaluation using the method in Study 6, and we find evidence for undervaluation, $M = 0.06$, 95% CI = [0.04, 0.08], $t(208) = 5.64$, $p < .001$. Choice share of M_{HL} was 6 percentage points lower than choice share of M_H . We tested for moderation by transparency by calculating the within-subject differences in undervaluation on high-transparency trials and undervaluation on low-transparency trials. Using a one-sample t-test we find significant evidence of the moderation ($M = -0.03$, 95% CI = [-0.06, -0.01], $t(208) = -2.76$, $p = .006$). There is less undervaluation when the M_H - M_L dominance is more transparent ($M(HT) = 0.04$, 95% CI = [0.02, 0.06], $t(208) = 3.70$, $p < .001$; $M(LT) = 0.07$, 95% CI = [0.05, 0.10], $t(208) = 5.95$, $p < .001$). This moderation indicated attenuation but not elimination. We still find evidence of undervaluation even when the dominant relationship is transparent. We also manipulated transparency in Studies 6d and 6e, but did not find a significant interaction in either study, so we do not strain to interpret the result here too much.

We find that 61% (127/209) of participants exhibited undervaluation overall (46% (97/209) exhibited undervaluation on high-transparency trials and 59% (123/209) exhibited undervaluation on low-transparency trials). Of the remaining 39%, 10% (20/209) exhibited no effect and 29% (62/209) exhibited an effect in the opposite direction. As in prior studies, we expect the percentage exhibiting undervaluation would increase with more choices. We observe

undervaluation for 9/10 of the choice sets overall, 8/10 of the high-transparency choice sets, and 9/10 of the low-transparency choice sets.

Attention to S. As specified in Hypothesis H4, we expect that choice of a multi-option alternative – and ultimately, undervaluation – to be associated with both alternative-level and sub-option-level attention.

We first test for a relationship between alternative-level attention and choice (H4a). In a trial-level analysis of the test trials using logistic regression, we regress choice of M_{HL} on the proportion of time spent on S (relative to M_{HL}), and we find a significant relationship between attention to an alternative and choice, $b = -7.20$, $SE = 0.48$, $p < .001$. Increased attention to S is associated with a lower probability of choosing M_{HL} .

In another test of alternative-level attention, we can examine whether alternative-level attention is related to choice at the aggregate level. For each subject, we computed the average difference in option-level attention between test and control trials (i.e., average proportion spent on S in S vs. M_{HL} choices – average proportion spent on S in S vs. M_H choices). We find a positive correlation between this average proportion mouse difference and the proportion of choices for M_{HL} , $b = -1.37$, $SE = 0.22$, $p < .001$.² Alternative-level attention also predicted undervaluation at the trial- and aggregate-levels.³

Attention to M_L . Next, we look to sub-option-level attention. H4b states that increased

² We also tested for trial-level associations between mouse movements and choice. Across all control, test, and trinary choices, we used logistic regression to regress choice (of S) on the proportion of mouse-hover time spent on S, with random intercepts and slopes at the subject level. We find a significant relationship between information acquisition and choice within-subject, $b = 4.79$, $SE = 0.18$, $p < .001$.

³ Trial-level attention to S (in test vs. control trials) is predictive of trial-level undervaluation ($b = 0.72$, $SE = 0.05$, $p < .001$). The difference in average proportion attention to S between test and control trials is predictive of aggregate undervaluation ($b = 0.77$, $SE = 0.19$, $p < .001$).

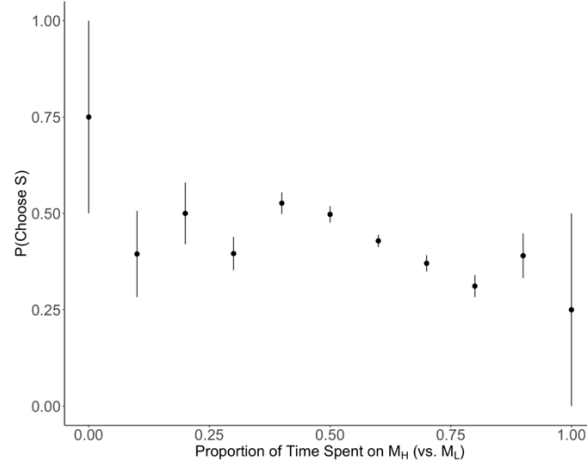
attention to an inferior sub-option will be associated with lower likelihood of choosing a multi-option alternative. In a trial-level analysis, we regressed choice for M_{HL} on the proportion of time spent on M_L relative to M_H (i.e., time spent on M_L / total time spent on M_{HL}). We find a significant negative relationship ($b = -1.71$, $SE = 0.25$, $p < .001$), which implies that an increase in the time spent looking at M_L (i.e., the inferior option in M_{HL}) is associated with a lower likelihood of choosing M_{HL} (i.e., a higher likelihood of choosing S) (figure S4). At the aggregate level, we find a similar result. A higher average proportion of time spent on M_L is associated with a lower proportion of choices for M_{HL} , $b = -0.78$, $SE = 0.19$, $p < .001$.⁴

A key difference between the two types of trials (control and test) is the number of stimuli on the screen. Not only do participants spend a smaller proportion of time on M_H in test choices (vs. control), they also spend a smaller proportion of time on M_H in test choices than they do in the equivalent trinary choices ($M = 0.01$, $t(208) = 3.16$, $p = .002$). Participants also choose M_{HL} less frequently in test choices than they do in trinary choices. All together, this supports the notion that attention directs and/or reflects the choice process.

Figure S4

Mouse-tracking and Choices

⁴ In a trial-level analyses of undervaluation, we see a significant relationship between attention to M_L and undervaluation, $b = 0.17$, $SE = 0.06$, $p = .001$. However, we do not see a significant relationship at the aggregate level, $b = -0.22$, $SE = 0.19$, $p = .24$). However, because it is impossible to measure attention to M_L in control choices, these metrics (especially at the aggregate level) are necessarily particularly noisy.



Notes: Relationship between mouse-tracking and choices in test choices. As participants spent more time on M_H (relative to M_L), they were less likely to choose the single-option alternative (S), $b = -1.71$, $SE = 0.25$, $p < .001$. 60% of all proportion of time spent on M_H lie in the interval $[0.5, 0.75]$. Within that interval, the coefficient is even more sharply negative (-3.05). Bars represent s.e.m. across participants.

An advantage of mouse-tracking (and other process tracing measures) is the ability to look further into the decision process than simple choice outcomes allow. The significant transparency moderation lends itself particularly well to investigation through the lens of process tracing. Thus, we compared the proportion of time spent on M_L (relative to M_H) in the high-transparency trials to the proportion of time spent on M_L (relative to M_H) in the low-transparency trials. We found a small but significant difference, such that subjects spent relatively less time on M_L when the dominance was more transparent, $M = -0.01$, 95% CI = $[0.002, 0.02]$, $t(207) = 2.46$, $p = 0.01$. Moreover, this subject-level difference in hover-times correlates with the subject-level difference in undervaluation between high- and low-transparency trials, $r = 0.13$, 95% CI = $[-0.008, 0.26]$, $t(206) = 1.86$, $p = .06$. In other words, as subjects spent less relative time on M_L in high-transparency trials (vs. low-transparency trials), they showed a smaller degree of undervaluation in high-transparency trials (vs. low-transparency trials).

We also compared choices in the test choices (S vs. M_{HL}) to choices in the trinary (S vs.

M_H vs. M_L) choices. Participants chose the multi-option alternative (M_{HL}) in test choices significantly less than they chose the same two options (M_H ; M_L) in the trinary choices, $M = -0.07$, 95% CI = $[-0.09, -0.05]$, $t(208) = -6.77$, $p < .001$ (figure S5). In this study, participants chose the multi-option alternative (M_{HL}) in test choices significantly less than they chose the higher-valued of the two (M_H) in trinary choice, $M = -0.02$, 95% CI = $[-0.05, -0.001]$, $t(208) = -2.09$, $p = .04$. Participants also chose M_H significantly more in control choices than in trinary choices, $M = 0.04$, 95% CI = $[0.02, 0.05]$, $t(208) = 3.62$, $p < .001$.

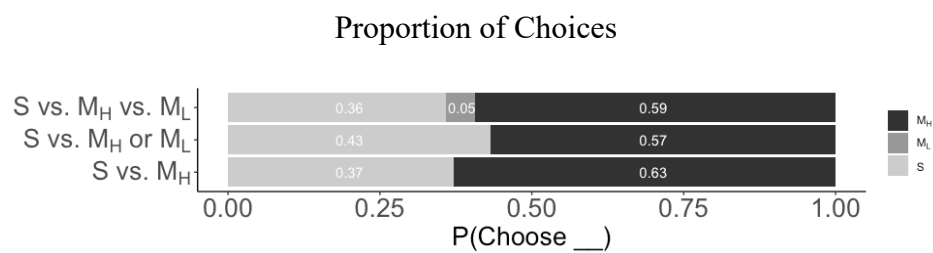
We examined mouse-tracking differences between trinary and multi-option alternative choices. First, we examined the difference between average proportions of time spent on S, M_H , and M_L in each trial type (i.e., trinary vs. test) and compared these differences to undervaluation. No correlations were significant (all $ps > 0.1$). Additionally, we looked at the difference in average proportion of transitions between M_H and M_L in the trinary trials compared to the test trials (i.e., number of transitions between M_H and M_L / total number of transitions). We did not find any significant correlations there either (all $ps > 0.1$).

Although our main effect of interest (undervaluation) is statistically significant, and the presence of undervaluation is widespread across participants, there is variability in the size of the effect across participants ($M = 0.06$, $SD = 0.15$). Some of this variability can be attributed to individual differences. Specifically, the degree of undervaluation in low-transparency trials is correlated with the degree of undervaluation in high-transparency trials, $r = 0.50$, 95% CI = $[-0.39, 0.59]$, $t(207) = 8.30$, $p < .001$. In other words, if a participant displayed a high degree of undervaluation on one half of the trials, they also displayed a high degree of undervaluation on the other half. This is consistent with individual differences in the tendency to exhibit undervaluation, rather than a uniform extent of undervaluation combined with pure noise in the

choice data. In Studies 5 and 6e, we attempted to identify an individual difference measure that was correlated with undervaluation, but did not find any.

Unlike Studies 1-3, there were a limited number of unique choices and little variability in the relative dominance of M_H over M_L in Study 6b across choice sets, so the test of H2 regarding difference in value had very low power and we do not discuss it here.

Figure S5



Notes: Proportion of choices for each option in binary, test, and trinary choices for Study 6b.

Appendix K: Study 6c

Study 6c provides a replication of our main effect in the domain of gambles, without measuring or manipulating attention.

Method

Participants. For this preregistered study, we collected responses from 304 AMT workers. They earned \$1.75 (the first 20 participants) or \$1.90 (the remaining 284 participants) for their participation. Five randomly-selected participants also received the outcome of one of their decisions, as detailed below.

Materials and Procedure. This study was similar to Study 1, with the following changes. First, choice options were gambles, as depicted in figure 1. As in Study 6, these gambles were drawn from a large pool of potential gambles from various combinations of drawing cards of different suits, flipping coins or pairs of coins, or rolling numbers on a six-sided die. Control choices, test choices, and trinary choices (described below) were matched on winning dollar amounts and probabilities but could vary (non-systematically) in terms of the specific mechanism as shown in figure 1. Second, participants did not rate any options, as we could model each option's value in terms of its payout and probability. Third, in addition to test choices (e.g., choosing between S and a multi-option alternative M_{HL}) and control choices (e.g., choosing between S and M_H), participants also made trinary choices (e.g., choosing among gambles S, M_H , and M_L). In total, participants made 34 incentivized choices (including three

attention check questions and one M_H vs. M_L choice from a randomly-selected test choice).

At the end of data collection, we randomly selected five participants and then randomly selected one of their choices to play out. Participants were aware of this structure. We opted to reward one trial instead of all of the trials to avoid stockpiling/strategy variability across trials (Juechems et al. 2017). In this study, three of the five participants won money (\$6, \$7, and \$10). See below for additional information on the trial generation process.

As in Studies 1-3, participants completed several comprehension check questions, including a question about the meaning of the test choices, i.e., that choosing M_{HL} implied that they would get to choose one option from the set M_{HL} . They were required to get all questions correct before moving on to the choices.

Trial Generation Process. To generate the 34 trials, we used a very similar process to Study 6. The only constraint we placed on the assignment of domains/probabilities to the three roles (S , M_H , and M_L) was to require that the probability of M_H be greater than the probability of M_L . If this was not possible (i.e., if the sampled probabilities were equal), then we resampled all three roles.

At the end of the survey, participants also completed a corresponding M_H vs. M_L trial from one of the 10 trial sets that they completed. Three attention check trials were used as detailed in Appendix H.

Exclusions and Data Preprocessing. As specified in our preregistration, we excluded anyone who picked an obviously dominated option in any of the three attention-check questions. This resulted in the exclusion of 63 participants, leaving us with a sample size of 241.

Results

Preregistered Results. To test for undervaluation, for every participant, we calculated $S > M_{HL}$ as (Chose S in test choices / Number of test choices). We also calculated $S > M_H$ as (Chose S in control choices / Number of control choices). We tested $(S > M_{HL}) - (S > M_H)$ using a one-sample t-test and found evidence for undervaluation, $M = 0.05$, 95% CI = [0.03, 0.08], $t(240) = 5.04$, $p < .001$. Choice of M_H was 5 percentage points higher than choice of M_{HL} (table 1). Using that approach, we find that 46% (111/241) of participants exhibited undervaluation overall, with 34% exhibiting no effect and 20% exhibiting an effect in the opposite direction. We would again expect the number exhibiting undervaluation to increase as the number of trials increases.

Because the probabilities of M_H and M_L were randomly selected, we can examine the role of M_H - M_L dominance (i.e., how much better M_H is than M_L) in undervaluation. We used Choice Difference as our outcome measure, defined as (Chose S in test choices – Chose S in control choices) at the set-level. We regressed Choice Difference on the expected value of S, the summed expected values of M_H and M_L , and the difference in expected values between M_H and M_L , clustering the standard errors at the subject level (eq. 2).

$$\text{Choice Difference}_{it} = \beta_0 + \beta_1 EV_{S_{it}} + \beta_2 (EV_{M_H} + EV_{M_L})_{it} + \beta_3 (EV_{M_H} - EV_{M_L})_{it} + \epsilon_{it} \quad (2)$$

The coefficient on the difference in expected values was in the expected direction, though it was not statistically significant, $b_3 = 0.011$, $SE = 0.008$, $p = .16$ (figure S6a).

Exploratory Results. One possible reason for the non-significant coefficient on expected

value is that participants were less attuned to the expected values of the options and more attuned to the probabilities themselves, perhaps due to how the gambles were displayed. To investigate this possibility, we turned to the control choices. Using logistic regression, we regressed choice of S on the expected values of S and M_H (EV), the monetary amounts associated with S and M_H (M), and the probabilities of S and M_H (P), clustering the standard errors at the subject level (eq. 3).

$$Choice_{it} = \beta_0 + \beta_1 EV_{S_{it}} + \beta_2 EV_{M_{H_{it}}} + \beta_3 M_{S_{it}} + \beta_4 M_{M_{H_{it}}} + \beta_5 P_{S_{it}} + \beta_6 P_{M_{H_{it}}} + \epsilon_{it} \quad (3)$$

After accounting for the effects of expected value (S: $b_1 = 0.42$, $SE = 0.12$, $p < .001$; M_H: $b_2 = -0.24$, $SE = 0.15$, $p = .11$), we find significant effects of both monetary amounts (representing the simple effects of monetary amount when probabilities are equal to 0; S: $b_3 = 0.19$, $SE = 0.06$, $p = .002$; M_H: $b_4 = -0.28$, $SE = 0.10$, $p = .003$) and both probabilities (representing the simple effects of probabilities when monetary amounts are equal to 0; S: $b_5 = 2.96$, $SE = 0.77$, $p < .001$; M_H: $b_6 = -2.83$, $SE = 0.91$, $p = .002$).

These results suggest that payout and probabilities contributed to evaluation of the gambles above and beyond their contribution to expected value. We thus adapted our preregistered analysis. Instead of regressing Choice Difference on expected values, we regressed it on probabilities.⁵ Specifically, we regressed Choice Difference on the probability of S, the summed probabilities of M_H and M_L, and the difference in probabilities between M_H and M_L, with clustered SEs at the subject level (eq. 4).

$$Choice\ Difference_{it} = \beta_0 + \beta_1 P_{S_{it}} + \beta_2 (P_{M_H} + P_{M_L})_{it} + \beta_3 (P_{M_H} - P_{M_L})_{it} \quad (4)$$

Consistent with our findings in Study 1, we find a marginally significant relationship between M_H-M_L dominance and Choice Difference ($b_3 = 0.11$, $SE = 0.06$, $p = .07$, figure S6b).

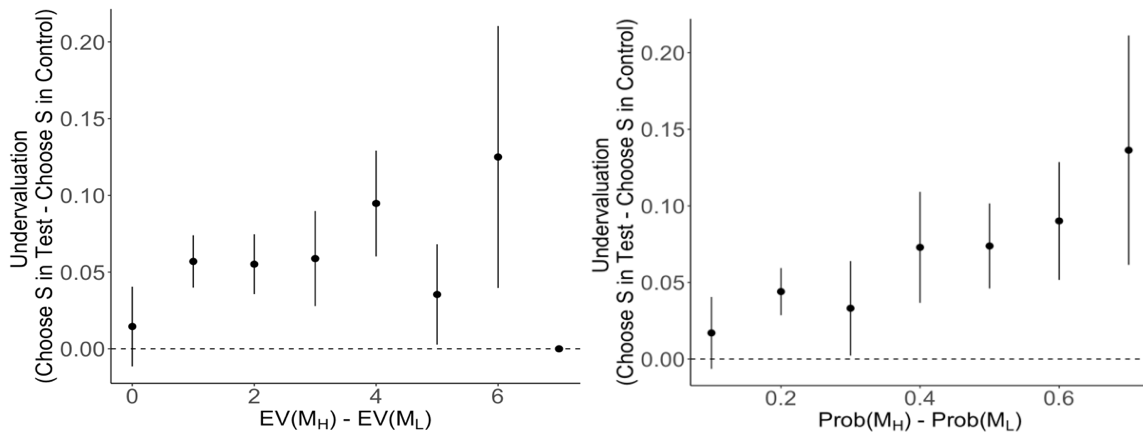
⁵ We are unable to conduct this analysis using monetary amounts because M_{M_H} was always the same as M_{M_L} .

This speaks in favor of the preference strength explanation (vs. measurement error), as probabilities here are known rather than measured with a noisy instrument.

We also compared choices in the test choices (S vs. M_{HL}) to choices in the trinary (S vs. M_H vs. M_L) choices. Using a one-sample t-test of (Choose M_{HL} in test choices) – (Choose M_H or M_L in trinary choices), we find that participants choose the multi-option alternative (M_{HL}) in test choices significantly less than they choose the same two options (M_H ; M_L) in the trinary choices, $M = -0.08$, 95% CI = $[-0.10, -0.06]$, $t(240) = -7.95$, $p < .001$ (see figure S8). There was not a significant difference between choice for the multi-option alternative (M_{HL}) in test choices and choice for the higher-valued option (M_H) in trinary choice, $M = 0.001$, 95% CI = $[-0.02, 0.03]$, $t(240) = 0.10$, $p = .92$. Participants chose M_H significantly more in control choices than they did in trinary choices ($M = 0.06$, 95% CI = $[0.04, 0.07]$, $t(240) = 5.79$, $p < .001$).

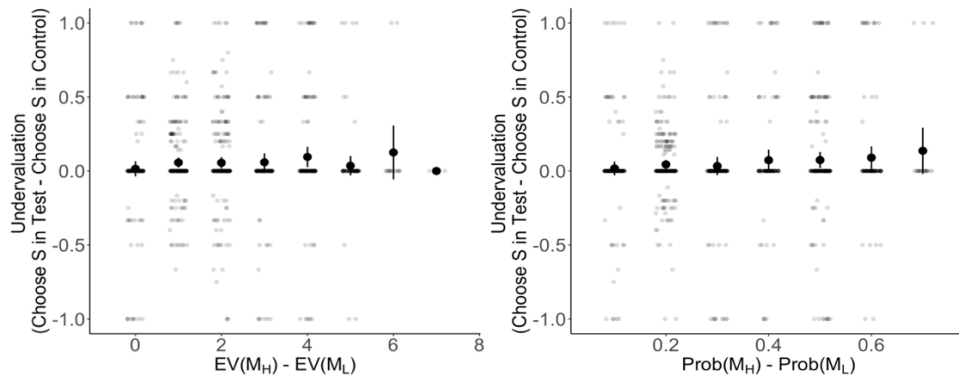
Figure S6

Dominance Strength and Undervaluation



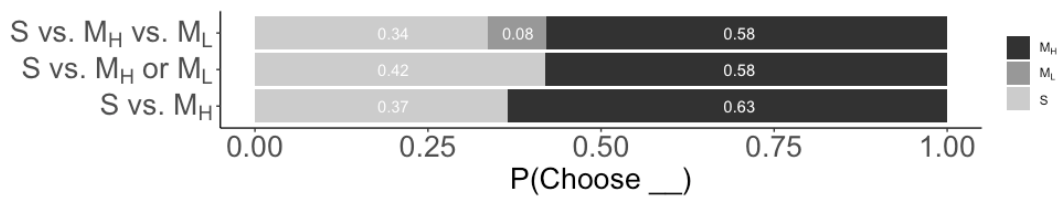
Notes: Relationship between ($M_H > M_L$) dominance strength and degree of undervaluation. (a) Dominance strength is defined using expected value. (b) Dominance strength is defined using probabilities. Across both definitions of dominance, undervaluation increases as M_H gets progressively better than M_L , though this relationship is not significant for expected value ($p = .16$) and marginally significant for probability ($p = .07$). Bars represent s.e.m. across participants. See figure S7 for plots with data overlaid.

Figure S7: Dominance Strength and Undervaluation



Notes: Relationship between ($M_H > M_L$) dominance strength and undervaluation. Across both definitions of dominance ((a) expected value, (b) probabilities), undervaluation increases as M_H gets progressively better than M_L , though this relationship is not significant for expected value ($p = .16$) and marginally significant for probability ($p = .07$). Bars indicate s.e.m. across participants.

Figure S8: Proportion of Choices



Notes: Proportion of choices for each option in binary, test, and trinary choices for Study 6c.

Appendix L: Study 6d

Method

Participants. For this preregistered experiment, we collected responses from 298 Amazon Mechanical Turk workers. They earned \$2.50 (the first 20 participants) or \$1.50 (the remaining 278 participants) for their participation. Five randomly-selected participants received the outcome of one of their decisions, as detailed below.

Materials and Procedure. This study was very similar to Study 6c, except for the following changes. First, the choices were not randomly generated as in Study 6c. See below for trial generation information. The other important difference is that the M_H vs. M_L dominance relationship was designed to be either high-transparency or low-transparency as in Study 6b. Participants made a total of 39 or 40 choices. At the end of the study, we randomly selected five participants, but none of them won anything from their randomly-selected decision.

Choices. For each of 10 trial sets t (as specified in Study 6b) a given participant completed the following trials: (1) low transparency: S^{t1} vs. M_{HL}^{t1} (L), where M_H^{t1} and M_L^{t1} use different mechanisms; (2) high transparency: S^{t2} vs. M_{HL}^{t2} , where M_H^{t2} and M_L^{t2} use the same mechanism; and (3) control 1: binary choice between S^{t3} and M_H^{t3} (L), where the mechanism (card draw, die roll, or coin flip) for M_H^{t3} is the same as the mechanism for M_H^{t1} (L). If the mechanisms for M_H^{t1} (L) and M_H^{t2} (H) were different, then the participant completed a 4th trial for that set: (4) control 2: S^{t3} vs. M_H^{t3} (H), where the mechanism for M_H^{t3} (H) is the same as the

mechanism for M_H^{t2} (H). Across participants, the assignment of varieties within each trial set (i.e., whether the i in S^{ti} is 1 or 2 or 3 for each of the four types of choices) was randomized. Additionally, participants were assigned to one of two between-subjects presentation conditions; the odd (even) numbered sets presented the M_{HL} option as “ M_H or M_L ” and the even (odd) numbered sets presented the M_{HL} option as “ M_L or M_H ” to control for presentation-order effects. In one case where S^{t3} was identical to M_H^{t3} , the appropriate control trial was excluded (as we could use its noiseless expected value of 0.5 instead).

In addition to the main trials of interest, participants also completed three attention-check questions, in which one option was designed to be (trivially obviously) dominant over the other according to monetary winnings, probability, or both. These three trials were spaced so that they occurred roughly one-fourth, one-half, and three-fourths of the way through the survey. All of the main trials of interest were presented in a fully-randomized order.

At the end of the survey, participants also completed the two corresponding M_H vs. M_L trials (i.e., M_H^{t1} vs. M_L^{t1} and M_H^{t2} vs. M_L^{t2}) from one of the 10 trial sets.

Exclusions and Data Preprocessing. As specified in our preregistration, we excluded anyone who picked a dominated option in any of the three attention-check questions. This resulted in the exclusion of 104 participants, leaving us with a sample size of 194.

Results

Results (Preregistered). We analyzed the results as in Study 6 and found undervaluation, $M = 0.06$, 95% CI = [0.03, 0.09], $t(193) = 4.28$, $p < .001$. We tested for an effect of transparency as in Study 6b and did not find one, $M = 0.002$, 95% CI = [-0.02, 0.02], $t(193) = 0.12$, $p = 0.91$.

Using that approach, we find that 49% of participants exhibited undervaluation, 20% showed no difference, and 31% showed an effect in the opposite direction. On the high-transparency trials, 48% exhibited undervaluation, 21% exhibited no difference, and 30% showed an effect in the opposite direction. On the low-transparency trials, 51% exhibited undervaluation, 19% showed no difference, and 30% showed an effect in the opposite direction. At the choice set level, we find that all 10 of the choice sets exhibit undervaluation, 9/10 exhibit undervaluation on high-transparency trials, and all 10 exhibit undervaluation on low-transparency trials.

Exploratory Results. We tested for within-subject consistency in undervaluation by correlating the undervaluation in high-transparency trials with the undervaluation in low-transparency trials. We found a significant positive relationship, $r = 0.67$, $t(192) = 12.46$, $p < .001$. Participants who displayed more undervaluation in high-transparency trials also displayed more undervaluation in low-transparency trials.

Appendix M: Study 6e

Introduction

In Study 6e, we investigate possible connections between undervaluation and various individual difference measures: risk aversion, analytic-holistic thinking (Choi et al. 2007), and elaboration on potential outcomes (Nenkov et al. 2008). We predicted that greater undervaluation would be associated with more holistic thinking (i.e., treating the M_{HL} option as a holistic unit, rather than as its component parts) and less elaboration on potential outcomes.

In addition to investigating individual differences, we also sought to rule out some alternative explanations for the effect. First, it is possible that people choose S more when it is paired with M_{HL} because people have an absolute avoidance of multi-option alternatives. If this were the case, we would expect them to be more likely to choose M_L (over S) than M_{HL} .

Method

Participants. For this preregistered experiment, we collected responses from 298 Amazon Mechanical Turk workers. They earned \$2.50 for their participation. Five randomly-selected participants received the outcome of one of their decisions, as detailed below.

Materials and Procedure. This experiment was very similar to Study 6d, with the following changes. Participants completed S vs. M_L choices and trinary choices in addition to binary and multi-option choices. Moreover, at the end of the experiment, participants completed

5 incentivized risk aversion trials (Holt and Laury 2002). They also completed the analytic-holistic thinking scale (Choi et al. 2007) and the elaboration on potential outcomes scale (Nenkov et al. 2008).

Additional Trial Details. In this experiment, participants made 43 incentivized choices (including 3 attention check questions and 5 risk aversion measures); some were control binary choices (e.g., S vs. M_H), some were trinary choices (e.g., S vs. M_H vs. M_L) and some were test choices (e.g., S vs. M_{HL}). For each of 10 trial sets t , a given participant completed the following trials: (1) low transparency: S^{t1} vs. M_{HL}^{t1} (L), where M_H^{t1} and M_L^{t1} use different mechanisms; (2) high transparency: S^{t2} vs. M_{HL}^{t2} (H), where M_H^{t2} and M_L^{t2} use the same mechanism; (3) trinary: S^{t3} vs. M_H^{t3} vs. M_L^{t3} ; (4) SM_L : S^{t1} vs. M_L^{t1} (or S^{t2} vs. M_L^{t2}); (5) $M_H M_L$: M_H^{t2} vs. M_L^{t2} (or M_H^{t1} vs. M_L^{t1}); (6) control 1: binary choice between S^{t3} and M_H^{t3} (L), where the mechanism (card draw, die roll, or coin flip) for M_H^{t3} (L) is the same as the mechanism for M_H^{t1} (L). If the mechanisms for M_H^{t1} (L) and M_H^{t2} (H) were different, then the participant completed a seventh trial for that set: (7) control 2: S^{t3} vs. M_H^{t3} (H), where the mechanism for M_H^{t3} (H) is the same as the mechanism for M_H^{t2} (H).

Participants were assigned to either complete trial sets 1-5 or trial sets 6-10. For trial types 3 and 4 (i.e., trinary and S vs. M_L), participants were randomly assigned in each set to complete the first or second versions of the trials, as detailed above. The version for M_{HL} was set to be the opposite of the version participants completed for SM_L . The full list of possible trials is available in our Research Box.

At the end of the experiment, participants completed the two corresponding M_H vs. M_L trials from one of the five trial sets that they completed. They also completed 5 risk aversion trials (Holt and Laury 2002). Finally, they completed the analytic-holistic thinking scale (Choi et

al. 2007) and the elaboration on potential outcomes scale (Nenkov et al. 2008). As in Studies 6-6d, at the end of data collection, we randomly selected five participants and then randomly selected one of their choices to play out. In this experiment, two of the five participants won money (\$2 and \$10).

Exclusions and Data Preprocessing. As specified in our preregistration, we excluded anyone who picked the dominated option in any of the three attention-check questions. This resulted in the exclusion of 122 participants, leaving us with a sample size of 176.

Results

Preregistered Results. We tested for undervaluation using the approach in Study 6d, and find evidence for undervaluation, $M = 0.06$, 95% CI = [0.02, 0.09], $t(175) = 3.37$, $p < .001$. We tested for moderation by transparency as in Studies 6b and 6d and do not find evidence of the moderation, $M = 0.002$, 95% CI = [-0.04, 0.04], $t(175) = 0.11$, $p = 0.91$. 55% of participants exhibited undervaluation, 19% exhibited no difference, and 27% exhibited an effect in the opposite direction. 41% exhibited undervaluation on high-transparency trials, and 49% exhibited undervaluation on low-transparency trials. At the choice set level, 8/10 of the choice sets exhibit undervaluation overall, 8/10 exhibit undervaluation on high-transparency trials, and 8/10 exhibit undervaluation on low-transparency trials.

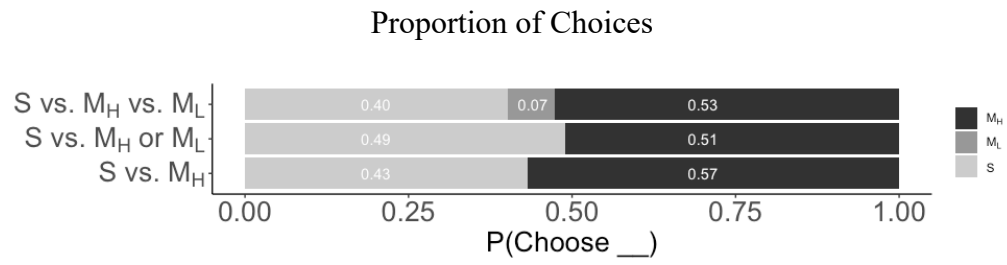
We did not find any evidence of a relationship between undervaluation and risk aversion ($r = 0.003$, $t(174) = 0.03$, $p = 0.97$) or elaboration on potential outcomes ($r = 0.007$, $t(174) = 0.10$, $p = 0.92$). We found a small negative correlation between undervaluation and

analytic/holistic thinking ($r = -0.17$, $t(174) = -2.28$, $p = 0.02$), which suggests that participants who reported being more holistic thinkers displayed less undervaluation. However, as this relationship is small, in the opposite direction as anticipated, and was one of several possible measured individual differences (increasing the probability of false positives), we do not strain to interpret it here.

With this data, we were also able to address two possible alternative explanations for our results. First, one possible explanation is that the undervaluation effect is due to blind avoidance of M_{HL} options. Comparing choice proportions in the multi-option alternative choices to choice proportions in the S vs. M_L choices allows us to address this possibility. People choose M_{HL} more often than they choose M_L , which indicates that they do not blindly avoid the M_{HL} options, $M = 0.18$, 95% CI = [0.15, 0.22], $t(175) = 10.93$, $p < .001$.

Exploratory Results. Another alternative explanation is that our results are simply due to noisy and/or careless responding. However, when we limit our dataset to participants who never chose a dominated option (i.e., chose M_H in all of the M_{HL} choices), the effect is still there and is even larger, $M = 0.11$, $t(91) = 5.57$, $p < .001$. Similarly, when we further restrict our dataset to participants who also got all of the comprehension questions correct on the first try, we find a similar (stronger) effect, $M = 0.10$, $t(67) = 4.61$, $p < .001$. Thus, when we limit our dataset to more attentive participants, the effect does not get smaller and, if anything, it gets larger.

Figure S9



Notes: Proportion of choices for each option in binary, test, and trinary choices for Study 6e.

Appendix N: Heterogeneity Simulation

To simulate behavior in multi-option alternative decisions, we used the data from Studies 1 and 1b. First, using all control choices from the data, we estimated the probability of choosing a film in a control choice, based on the rating of the film and the rating of its competitor (i.e., using logistic regression we regressed ChooseFilmM_H on RatingS and RatingM_H). Then, using all test choices from the data, we estimated the probability of choosing a multi-option alternative (i.e., M_{HL}), based on the ratings of the single-option alternative and both options of the multi-option alternative (i.e., using logistic regression we regressed $\text{ChooseM}_H\text{M}_L$ on RatingS , RatingM_H , and RatingM_L). We use these results to estimate the probabilities of choosing single-option and multi-option alternatives as detailed below.

Next, for each unique pair of films ($50 \times 49 / 2 = 1225$ pairs), we identified the more-popular film (i.e., the film that was rated higher by more participants, which we will call "Film A") and the less-popular film (i.e., whichever film is not Film A, which we will call "Film B"). For each unique pair of films for each participant, there is also a personally-preferred film (which we call "Film M_H ") and a personally-unpreferred film (which we call "Film M_L ").

We also identified (using participant-level ratings of S, M_H , and M_L and the estimates from the regressions identified above) at the participant level: (1) the probability that Film A would be chosen, (2) the probability that Film B would be chosen, (3) the probability that Film M_H would be chosen, (4) the probability that Film M_L would be chosen, and (5) the probability that a multi-option alternative comprising both films (Films A and B; a.k.a. Films M_H and M_L) would be chosen. We then averaged each of the five measures above across participants to get average probabilities for each pair of films.

Next, we considered the results assuming different levels of targeting capabilities. First, we consider the situation where we have no ability to target consumers based on their preferences. In this case including both films as options maximizes probability of choice. In other words, we expect "overvaluation" of the multi-option alternative, relative to the more-popular option (Figure 8). Second, we consider the situation where we are able to identify the personally-preferred film for a given consumer with some probability (w). In this case, we expect to observe probability (3) above in $w*100\%$ of consumers and observe probability (1) above in $(1-w)*100\%$ of consumers on average. To represent this in our simulations, we randomly selected which consumers' preferences would be accurately identified (using a binomial distribution with probability w). We repeated this random selection 1000 times, averaging at the movie-pair level. As w increases, and as offerings are tailored to individuals, expected undervaluation of the multi-option alternative increases as well.

Appendix O: Supplementary Table

Table S2: Summary of Choice Trials Across All Studies.

Study	Total Choices	Test	Control	Trinary	S vs. M _L	M _H vs. M _L	Attention Check Questions	Incentive-compatible	Attention	Other
1 1b 1c 1d	30	10	10	0	0	0	0			10 filler binary questions
2 2b	30	10	10	0	0	0	0			10 filler binary questions; post-choice ratings: S, M, M _H , M _L
3 3b	42	5	5	0	0	0	0			17 filler binary questions, 5 1v3, 5 2v2, 5 2v3
4	3 ^a	1	1				2			Between-subjects
5	21	5	5				1	X	X	10 filler questions, plus maximizing and need for cognitive closure
6	34	10	10			1	3	X	X	10 filler questions
6b	65	20	20	10	0	2	3	X	X	10 rank-order questions
6c	34	10	10	10	0	1	3	X		
6d	39-40	20	14-15	0	0	2	3	X		
6e	43	10	10	5	5	5	3	X		risk aversion, analytic / holistic thinking, elaboration on potential outcomes

^a Although participants made three choices, our main effect is based on only the first choice made (in a between-subjects design.)